

Math CS 121 HW 4

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1 (ND)

Question 1. 4.3.15:

A fair die is tossed successively. Let X denote the number of tosses until each of the six possible outcomes occurs at least once. Find the probability mass function of X .

Hint: For $1 \leq i \leq 6$, let E_i be the event that the outcome i does not occur during the first n tosses of the die. First calculate $\mathbb{P}(X > n)$ by writing the event $X > n$ in terms of $E_1, \dots, E_6$.

Proof. For each $n \in \mathbb{N}$, let E_i be the event that the number i doesn't occur during the first n tosses of the die. Since each toss of the die is independent, then for number i to not occur during the first n tosses, each toss must be the other 5 numbers, which has probability $\frac{5}{6}$ of happening for each toss. So, $\mathbb{P}(E_i) = (\frac{5}{6})^n$ (since each toss must be out of the 5 numbers, while every toss is independent).

Now, if consider the probability $\mathbb{P}(X > n)$,

□

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Question 2. *The demand of a certain weekly magazine at a newsstand is a random variable with probability mass function $p(i) = (10 - i)/18$, $i = 4, 5, 6, 7$. If the magazine sells for \$ a and costs \$ $2a/3$ to the owner, and the unsold magazines cannot be returned, how many magazines should be ordered every week to maximize the profit in the long run?*

Proof. Since $p(i) := \mathbb{P}(X = i)$ (while $i = 4, 5, 6, 7$), with i denotes the number of demand of the magazine, and each magazine sells for \$ a and costs \$ $2a/3$. Which, let k denotes the total number of magazines ordered. If selling out total of i magazines, the profit $Z_i = ia - \frac{2k}{3}a = \left(i - \frac{2k}{3}\right)a$. So, if fixing $k > 0$, together with the provided probability mass function, the expectation value of the profit is:

$$\mathbb{E}[Z] = \sum_{i=4}^7 p(i)Z_i = \frac{1}{3} \cdot \left(4 - \frac{2k}{3}\right)a + \frac{5}{18} \left(5 - \frac{2k}{3}\right)a + \frac{2}{9} \left(6 - \frac{2k}{3}\right)a + \frac{1}{6} \left(7 - \frac{2k}{3}\right)a \quad (1)$$

$$= a \left(\frac{47}{9} - \frac{2k}{3} \right) \quad (2)$$

□

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Question 3.

Proof.

□

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Question 4.

Proof.



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Question 5.

Proof.



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Question 6.

Proof.



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Question 7.

Proof.

□

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Question 8.

Proof.

□

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Question 9.

Proof.

□

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Question 10.

Proof.

